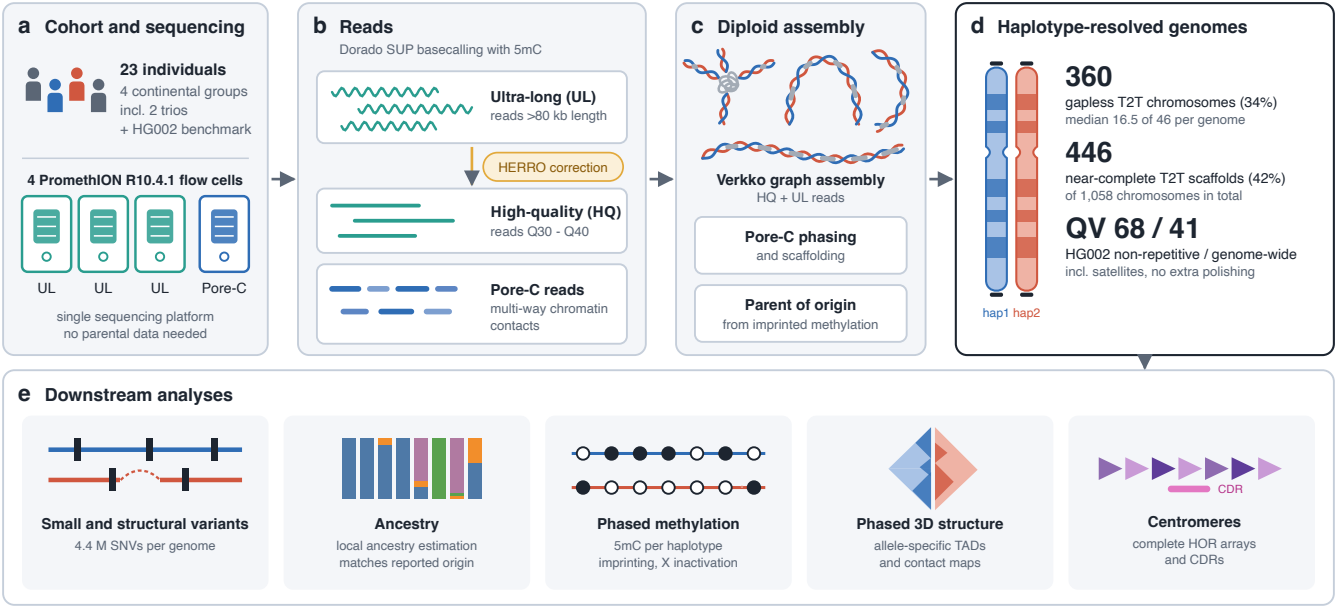

a Cohort and sequencing
23 individuals
4 continental groups
incl. 2 trios
+ HG002 benchmark
4 PromethION R10.4.1 flow cells
UL
UL
UL
Pore-C
single sequencing platform
no parental data needed
b Reads
Dorado SUP basecalling with 5mC
Ultra-long (UL)
reads >80 kb length
HERRO correction
High-quality (HQ)
reads Q30 - Q40
Pore-C reads
multi-way chromatin
contacts
c Diploid assembly
Verkko graph assembly
HQ + UL reads
Pore-C phasing
and scaffolding
Parent of origin
from imprinted methylation
d Haplotype-resolved genomes
hap1
hap2
360
gapless T2T chromosomes (34%)
median 16.5 of 46 per genome
446
near-complete T2T scaffolds (42%)
of 1,058 chromosomes in total
QV 68 / 41
HG002 non-repetitive / genome-wide
incl. satellites, no extra polishing
e Downstream analyses
Small and structural variants
4.4 M SNVs per genome
Ancestry
local ancestry estimation
matches reported origin
Phased methylation
5mC per haplotype
imprinting, X inactivation
Phased 3D structure
allele-specific TADs
and contact maps
CDR
Centromeres
complete HOR arrays
and CDRs